AS1109: Colour 3D Graphs for Extrema

Perspective surface plots (drawn in 2D PDF form)

This PDF collects the main functions whose local/global extrema were discussed in the le
teacher notes, and student notes. Each page shows a colourful 3D-style surface plot toge
its key critical/extreme points.

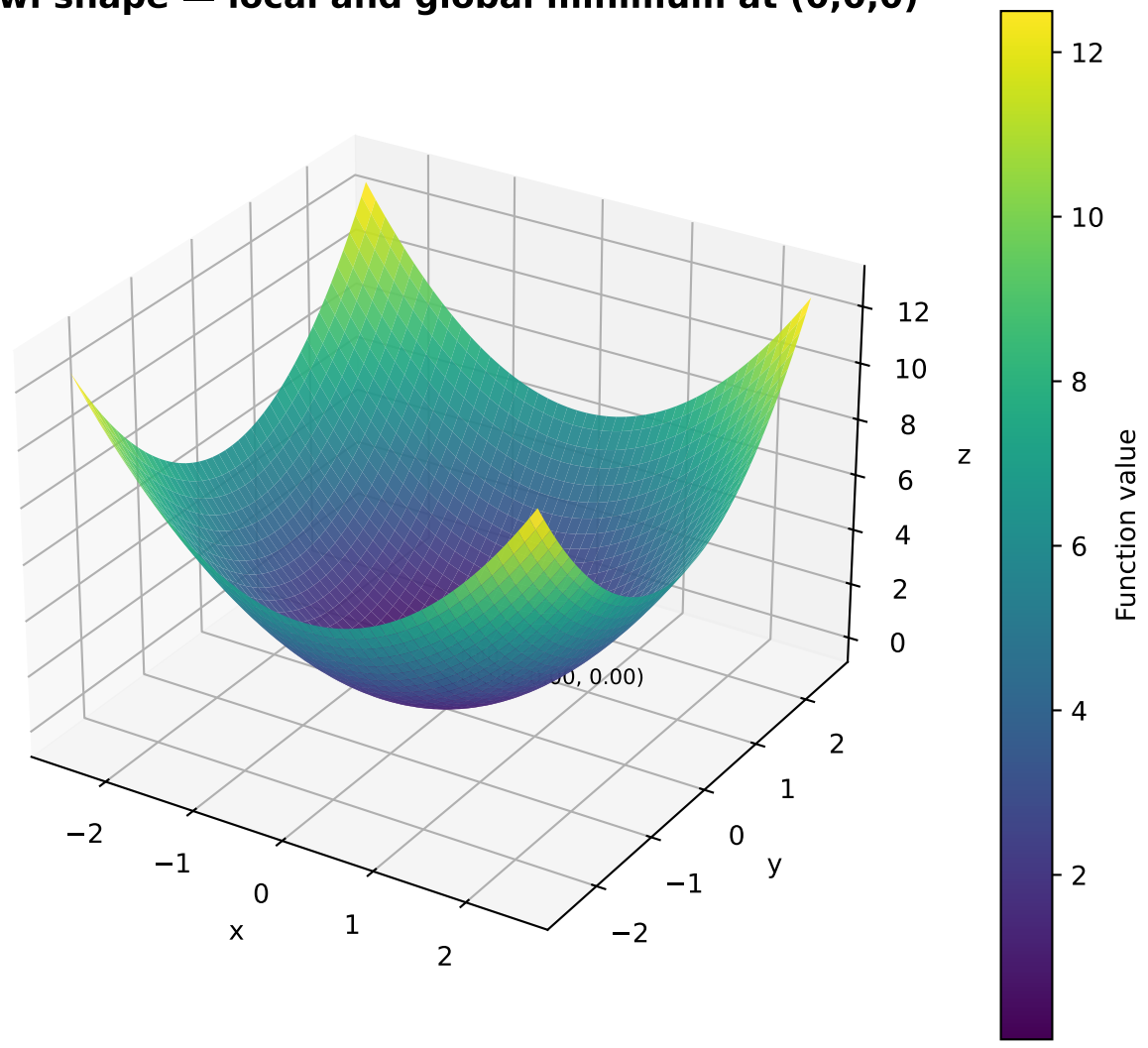
Included surfaces:
• bowl / cap / saddle prototypes
• the worked class example f(x,y)=x^3-3x+y^2
• the coupled quadratic Q=x^2+xy+y^2
• the D=0 counterexamples x^4+y^4 and x^4-y^4
• absolute-extrema-on-a-rectangle examples
• the idealised engineering models T, P, C, and E
• the restricted design-box example and a cap-shaped maximum example

Suggested classroom use:
1. show the page, 2. ask students to predict the type of point, 3. connect the picture with

Abbreviations used on the pages:
• min = minimum
• max = maximum
• abs = absolute/global
• local = neighbourhood-based

$$z = x^2 + y^2$$

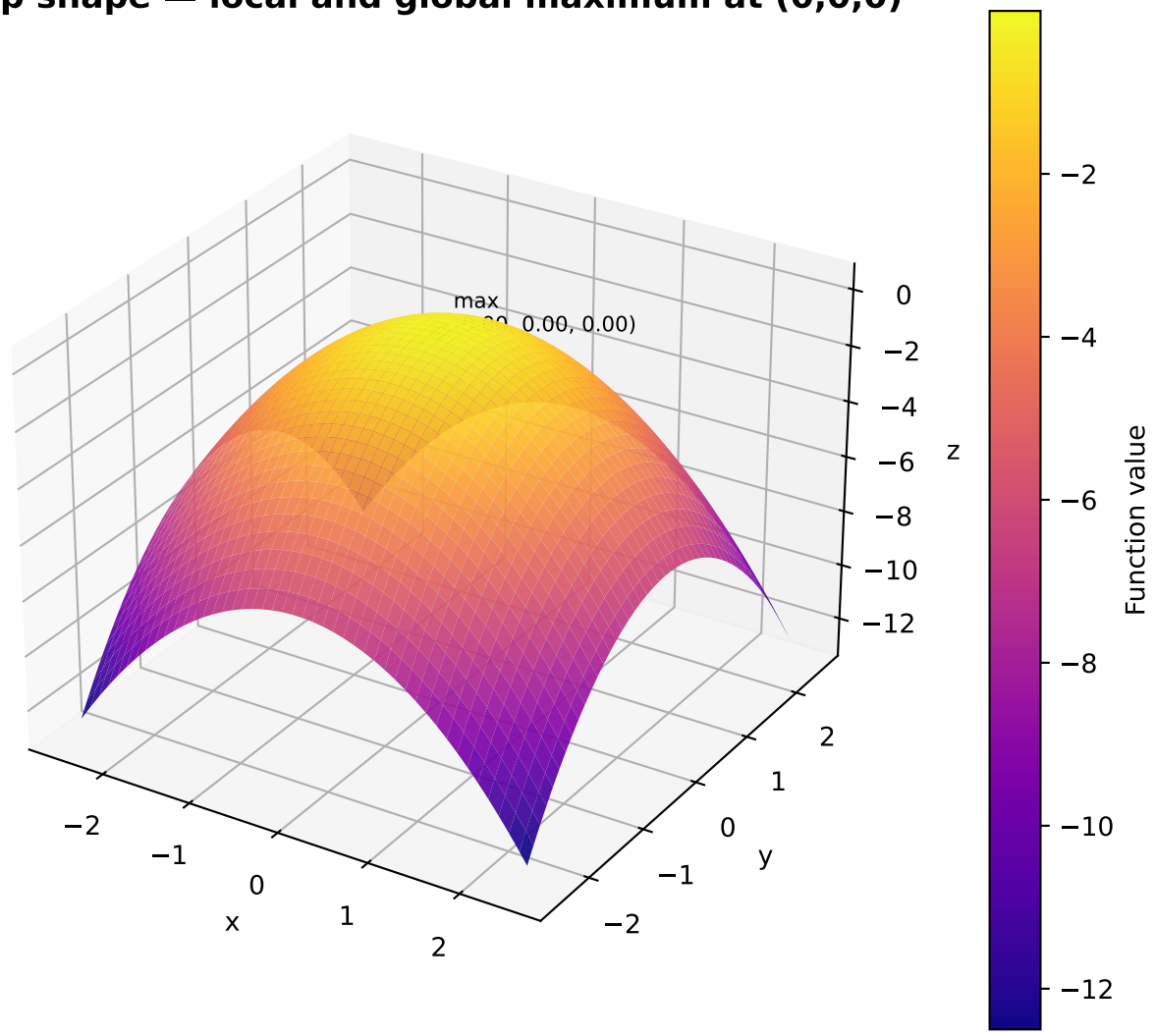
Bowl shape — local and global minimum at (0,0,0)



Key point(s): min
Bowl-shaped surfaces typically correspond to minima.

$$z = -x^2 - y^2$$

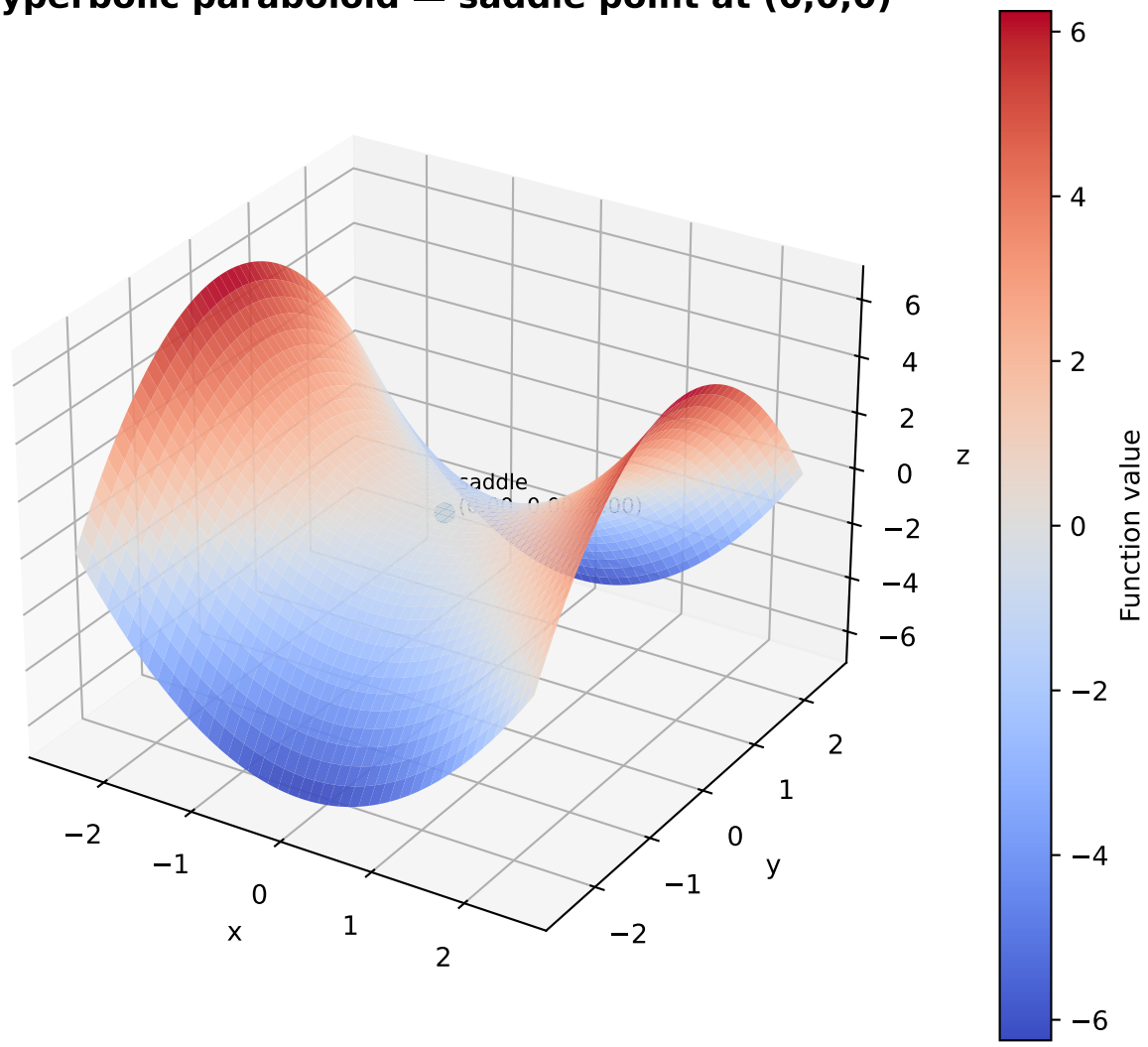
Cap shape — local and global maximum at (0,0,0)



Key point(s): max
A saddle goes up in some directions and down in others.

$z = x^2 - y^2$

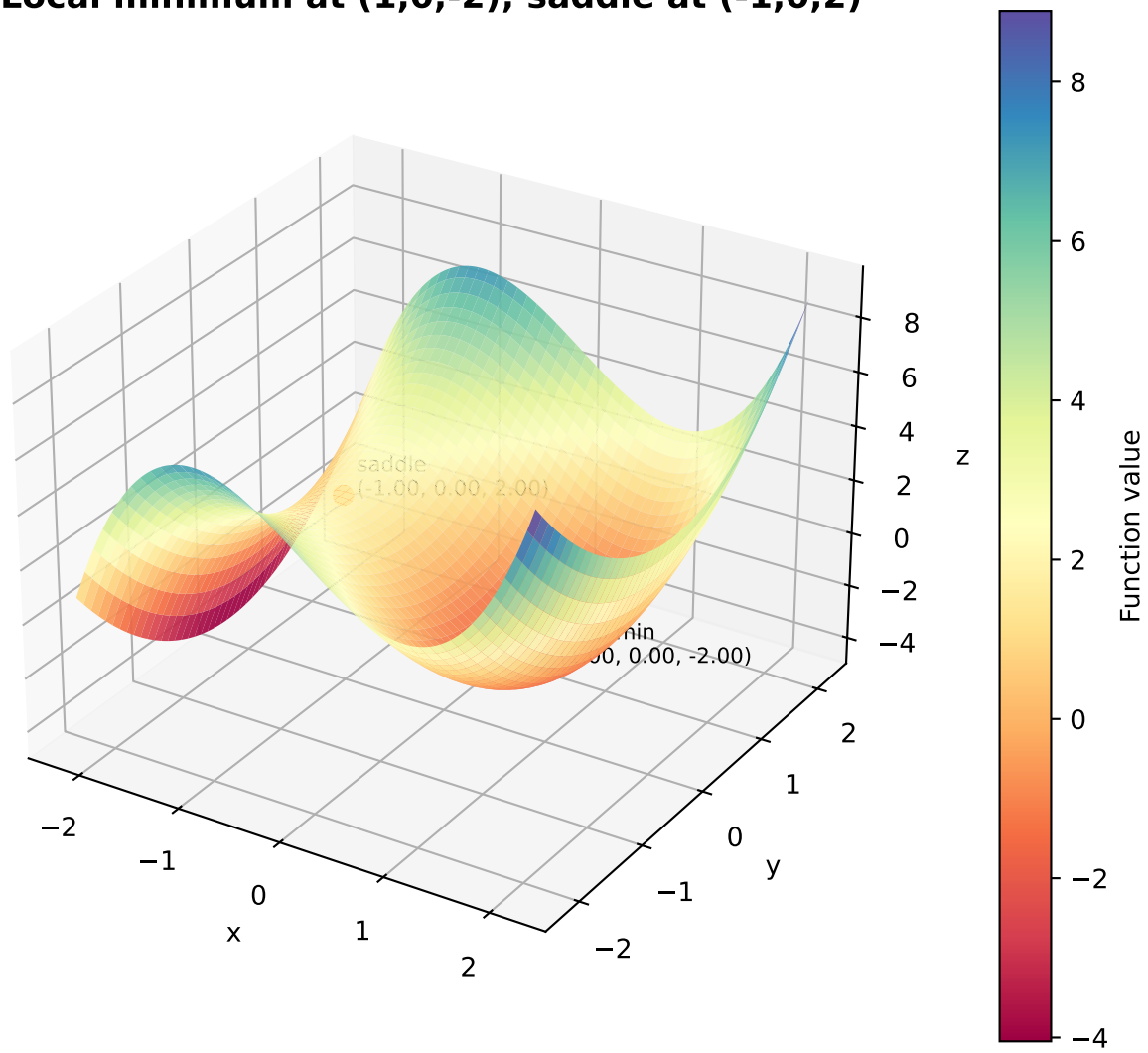
Hyperbolic paraboloid — saddle point at (0,0,0)



Key point(s): saddle
A saddle goes up in some directions and down in others.

$f(x, y) = x^3 - 3x + y^2$

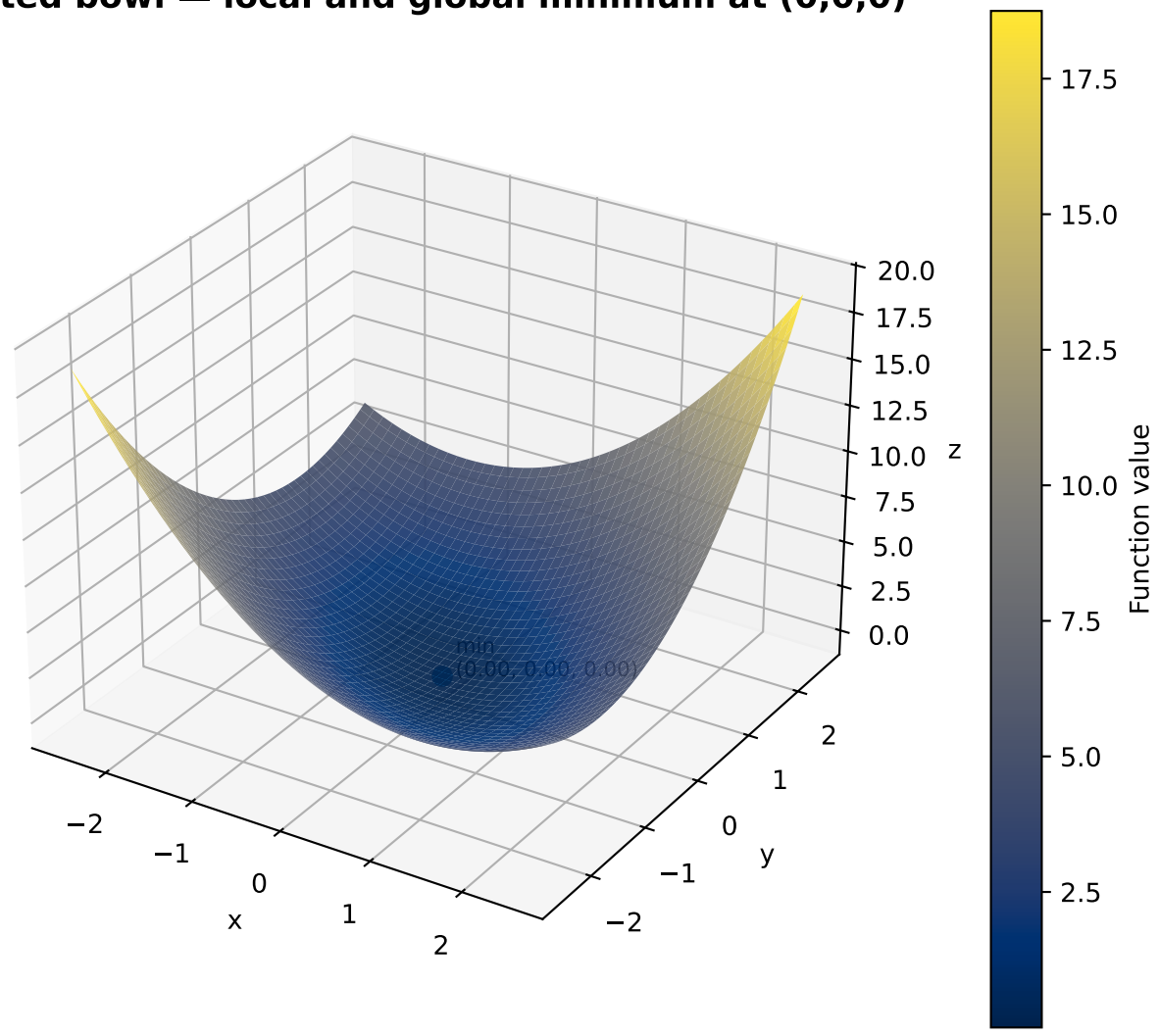
Local minimum at (1,0,-2); saddle at (-1,0,2)



Key point(s): local min, saddle
A saddle goes up in some directions and down in others.

$$Q(x, y) = x^2 + xy + y^2$$

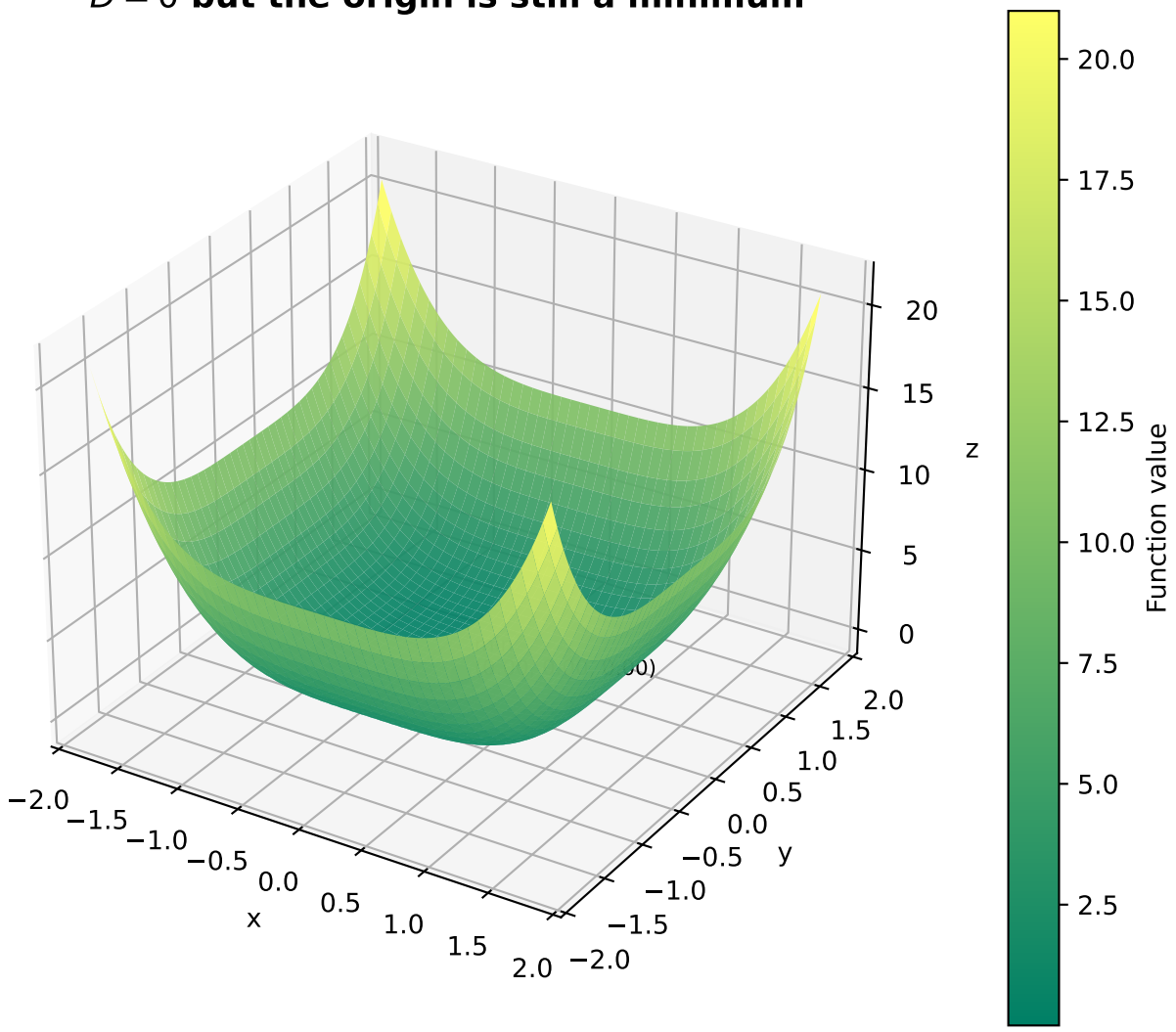
Tilted bowl — local and global minimum at (0,0,0)



Key point(s): min
Bowl-shaped surfaces typically correspond to minima.

$f(x, y) = x^4 + y^4$

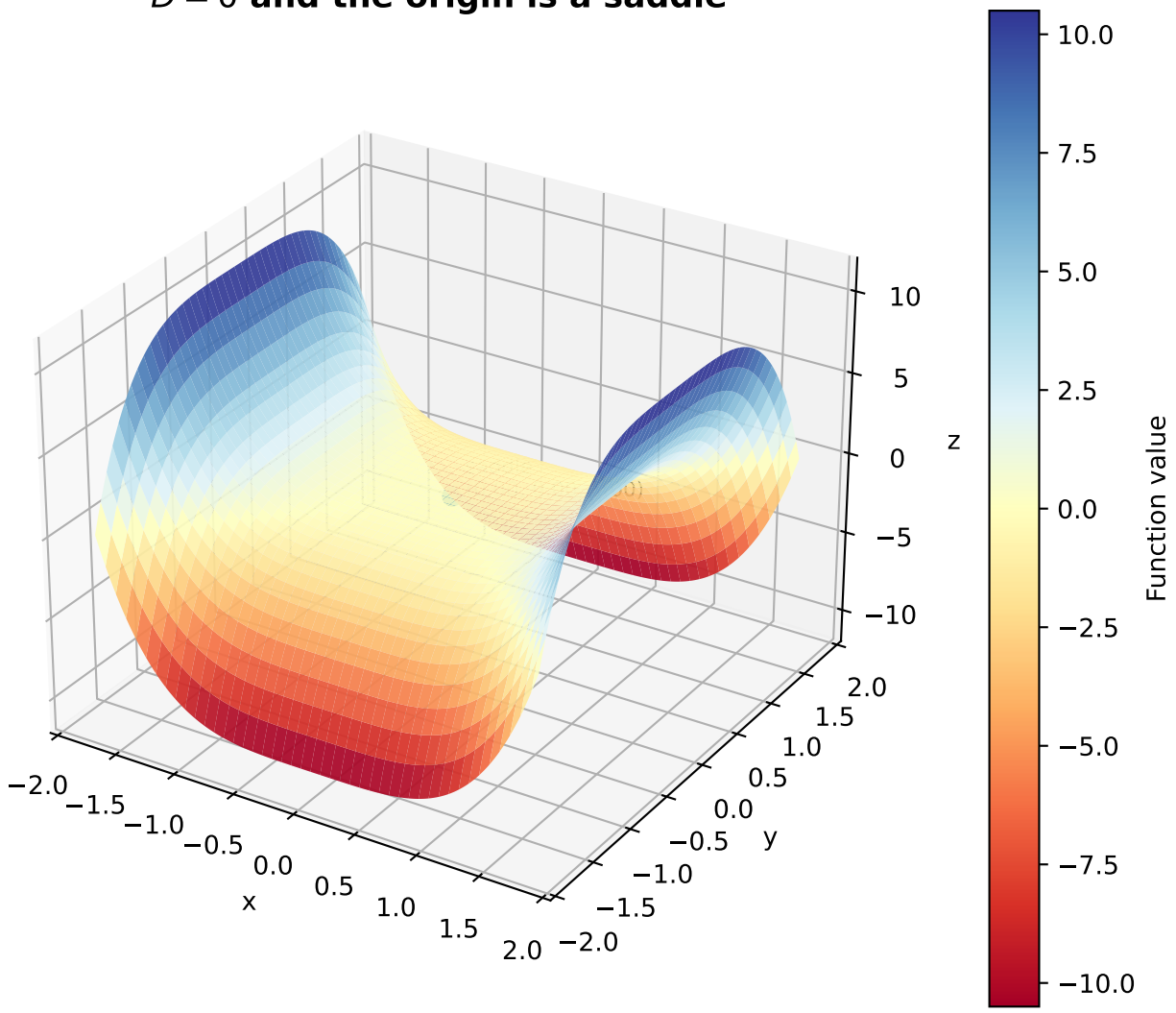
$D = 0$ **but the origin is still a minimum**



Key point(s): min

These pages illustrate why D = 0 is inconclusive.

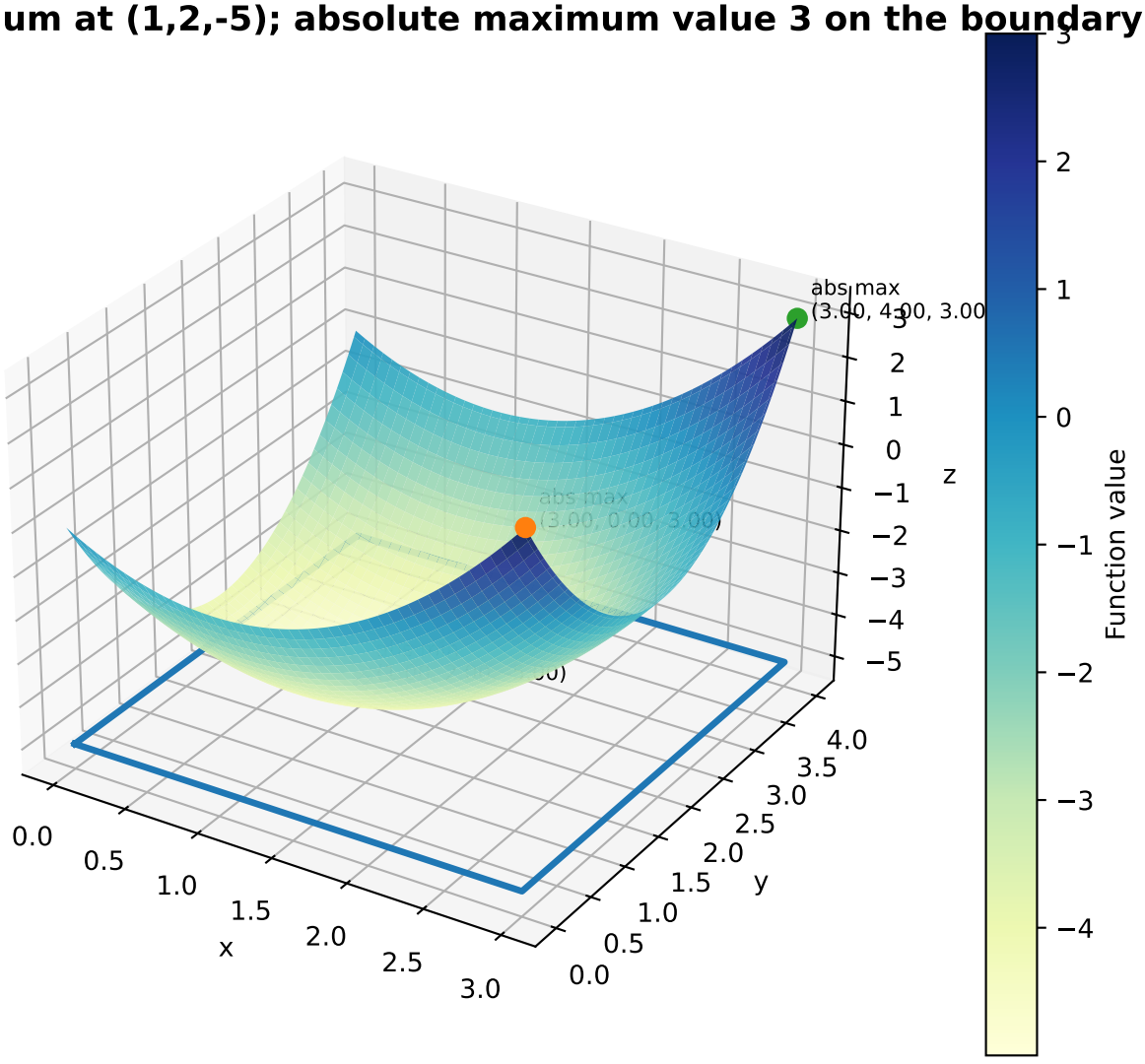
$g(x, y) = x^4 - y^4$
$D = 0$ **and the origin is a saddle**



Key point(s): saddle
These pages illustrate why D = 0 is inconclusive.

$f(x, y) = x^2 + y^2 - 2x - 4y$ **on** $0 \le x \le 3,\ 0 \le y \le 4$

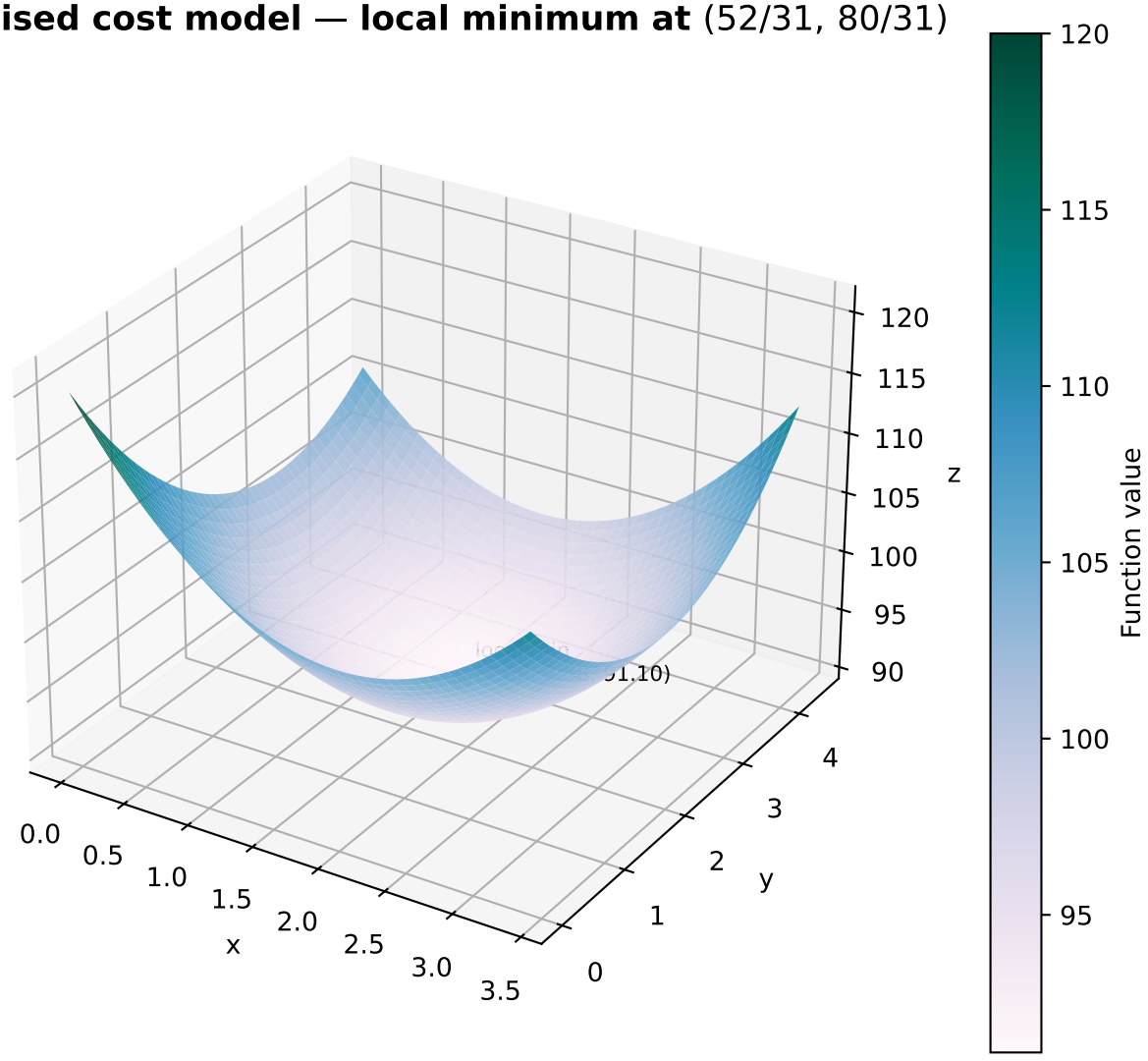
solute minimum at (1,2,-5); absolute maximum value 3 on the boundary



Key point(s): abs min, abs max, abs max
Cap-shaped surfaces typically correspond to maxima.

$$C(x, y) = 120 + 4x^2 + 2y^2 - 16x - 12y + xy$$

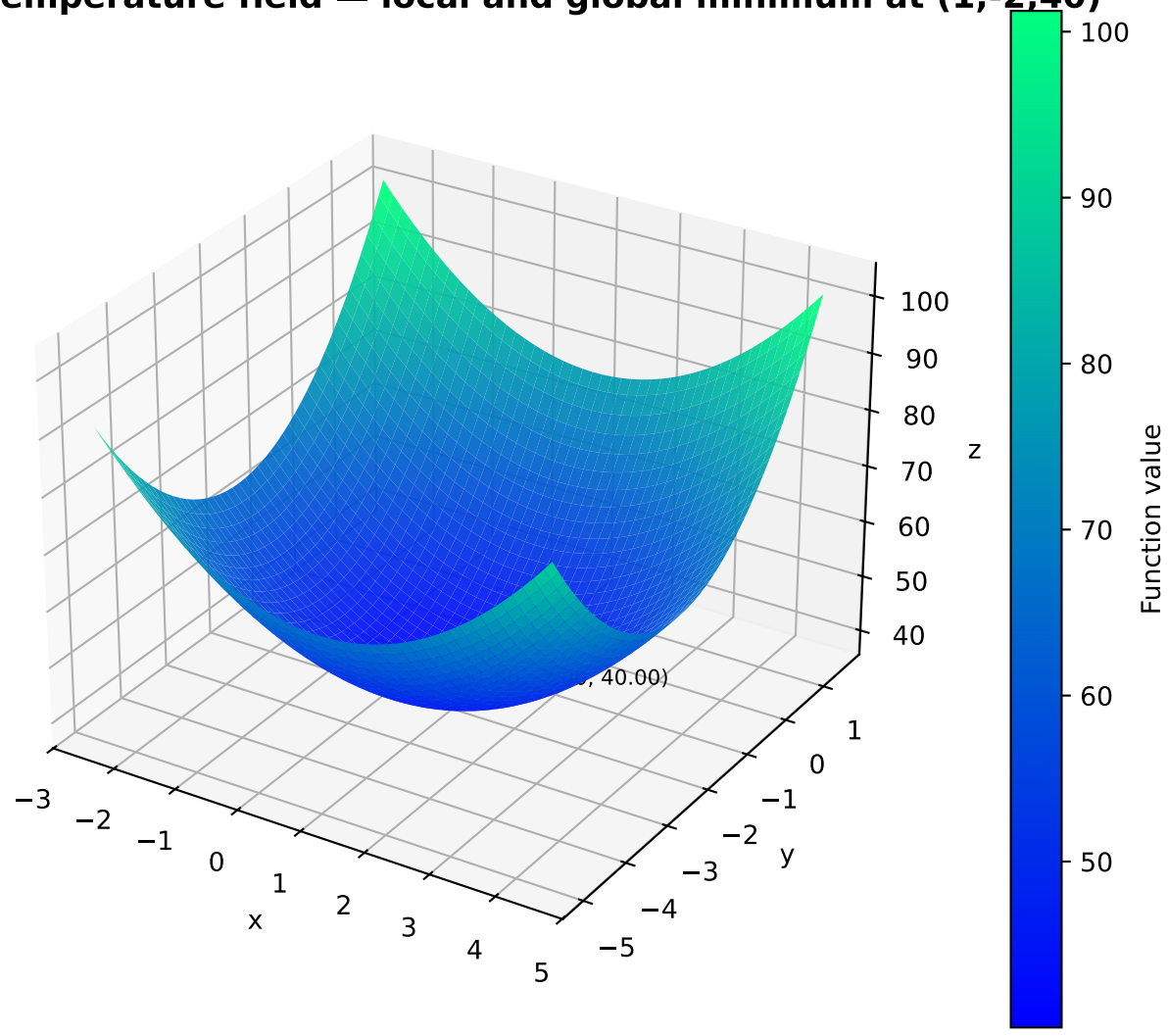
Idealised cost model — local minimum at (52/31, 80/31)



Key point(s): local min
Bowl-shaped surfaces typically correspond to minima.

$$T(x, y) = 40 + 2(x-1)^2 + 3(y+2)^2$$

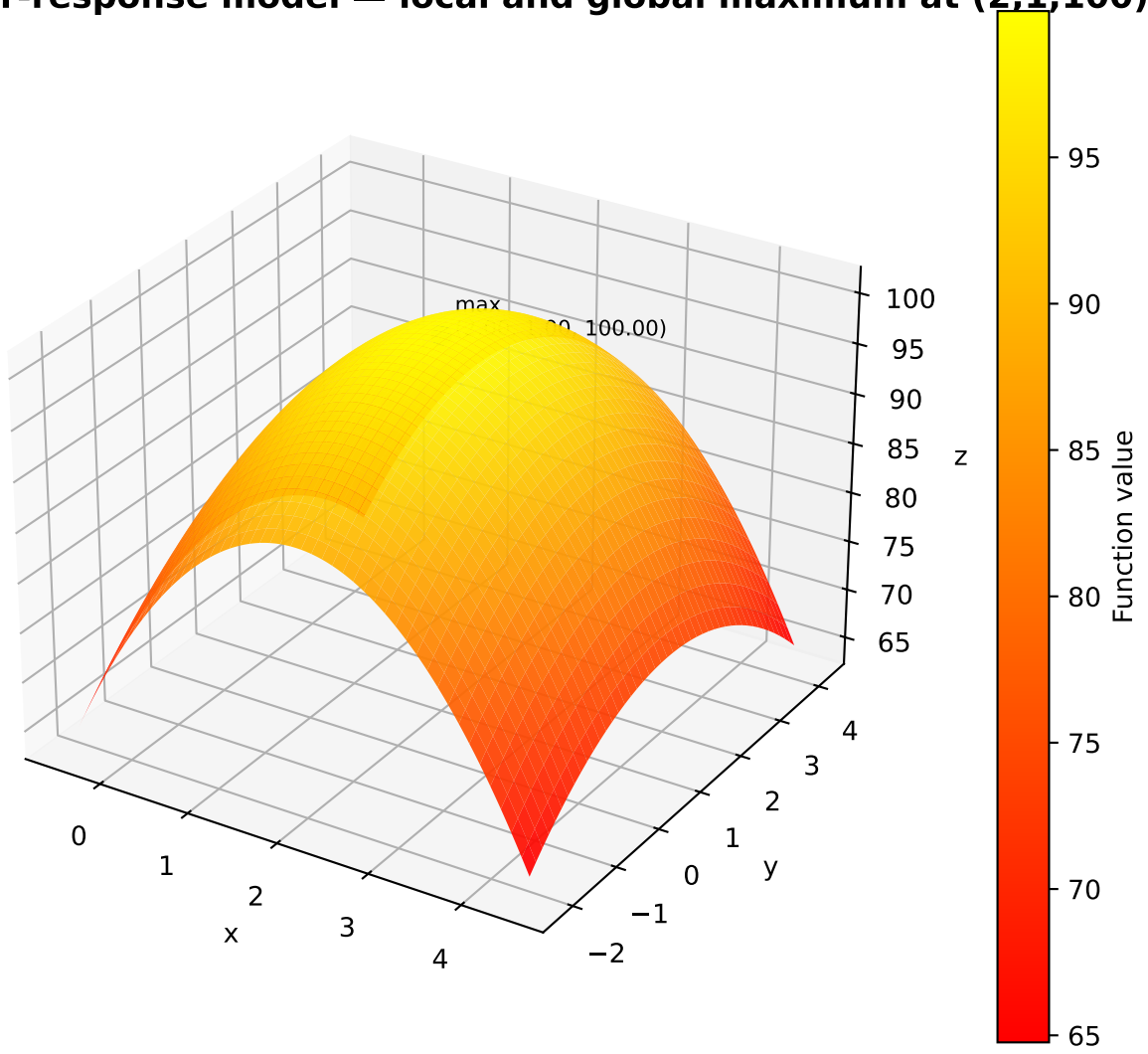
Idealised temperature field — local and global minimum at (1,-2,40)



Key point(s): min
Bowl-shaped surfaces typically correspond to minima.

$$P(x, y) = 100 - 4(x-2)^2 - (y-1)^2$$

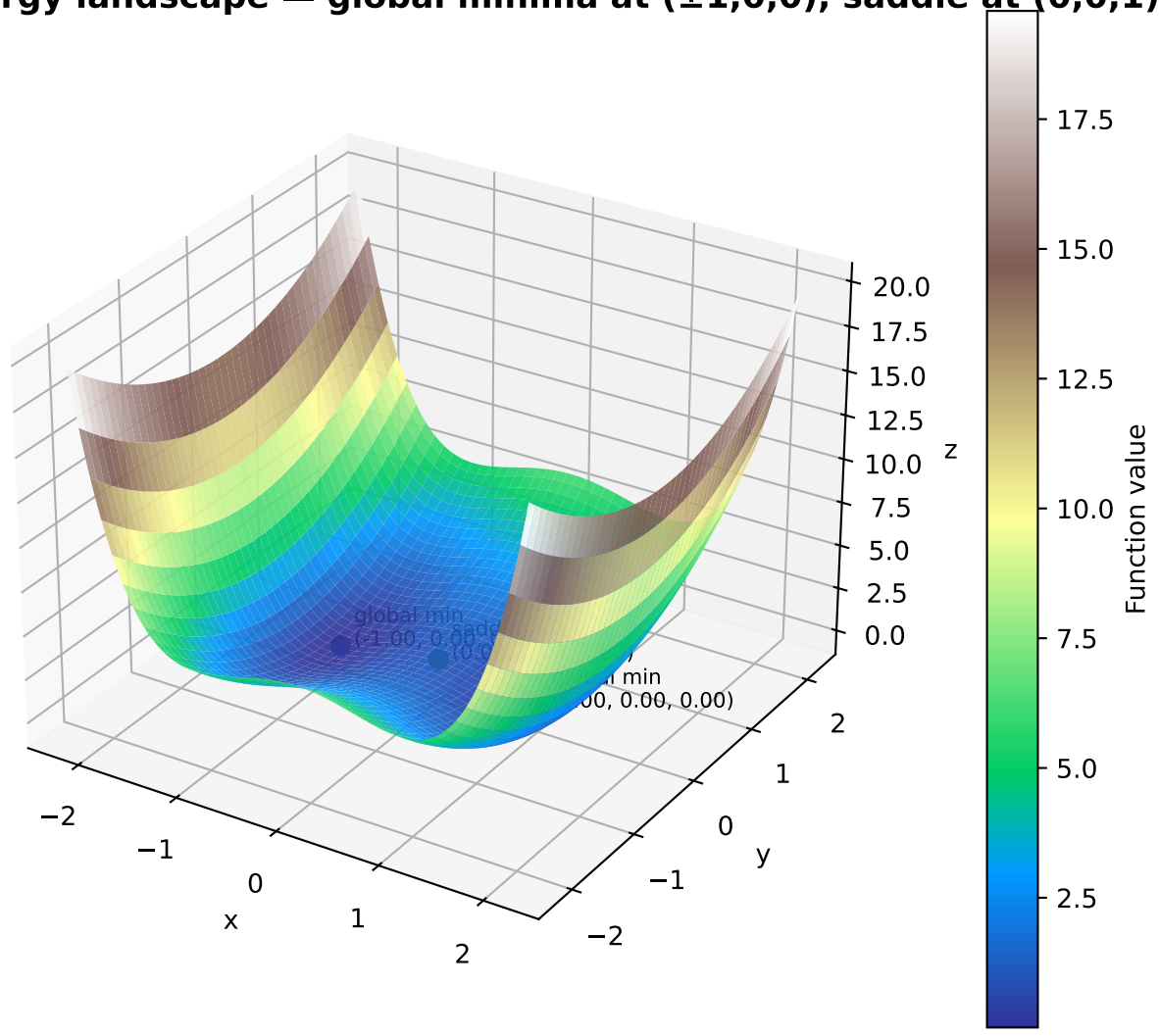
ealised power-response model — local and global maximum at (2,1,100)



Key point(s): max
Cap-shaped surfaces typically correspond to maxima.

$$E(x, y) = (x^2 - 1)^2 + y^2$$

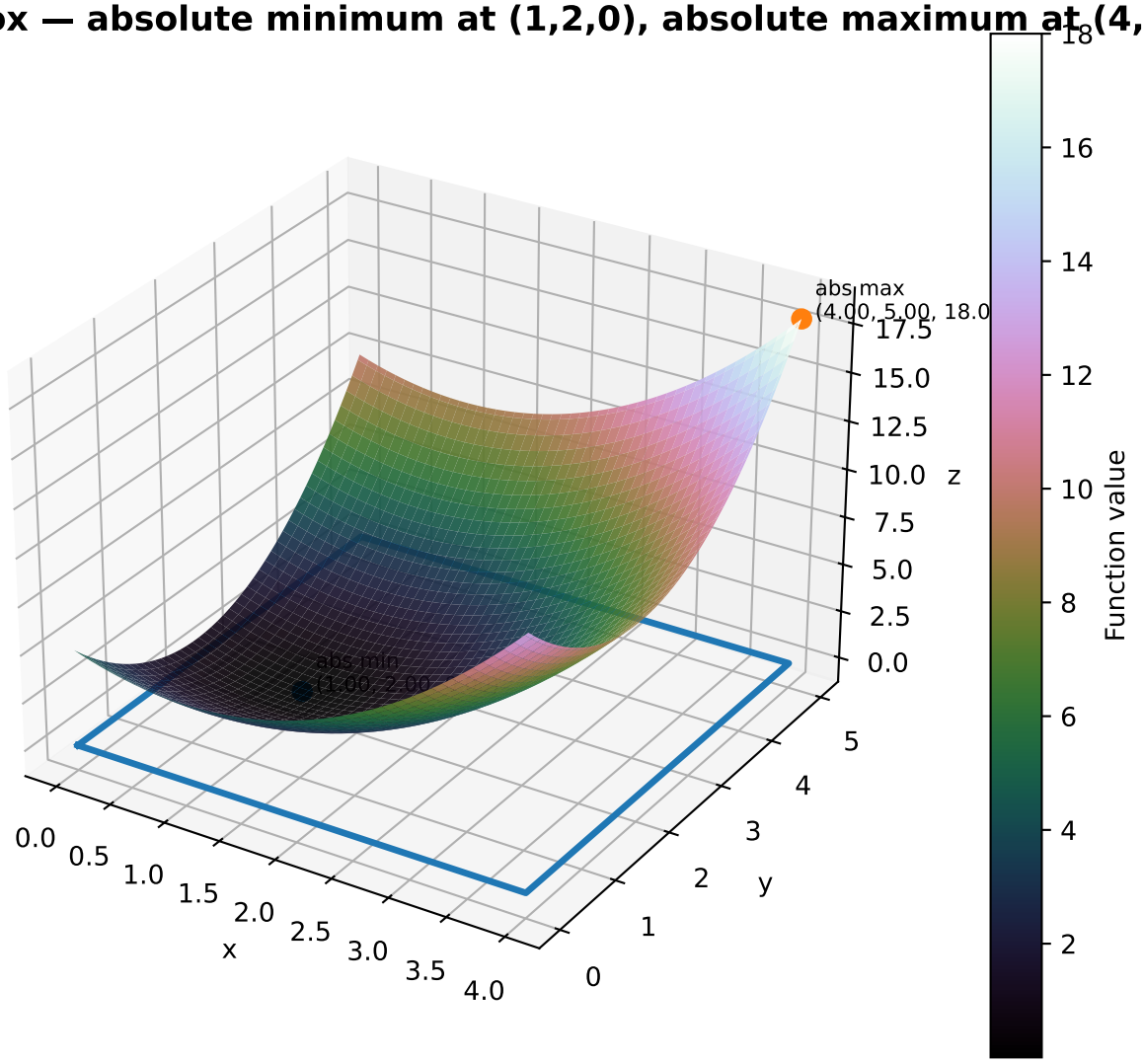
ble-well energy landscape — global minima at (±1,0,0), saddle at (0,0,1)



Key point(s): global min, global min, saddle
A saddle goes up in some directions and down in others.

$C(x, y) = (x-1)^2 + (y-2)^2$ **on** $0 \le x \le 4,\ 0 \le y \le 5$

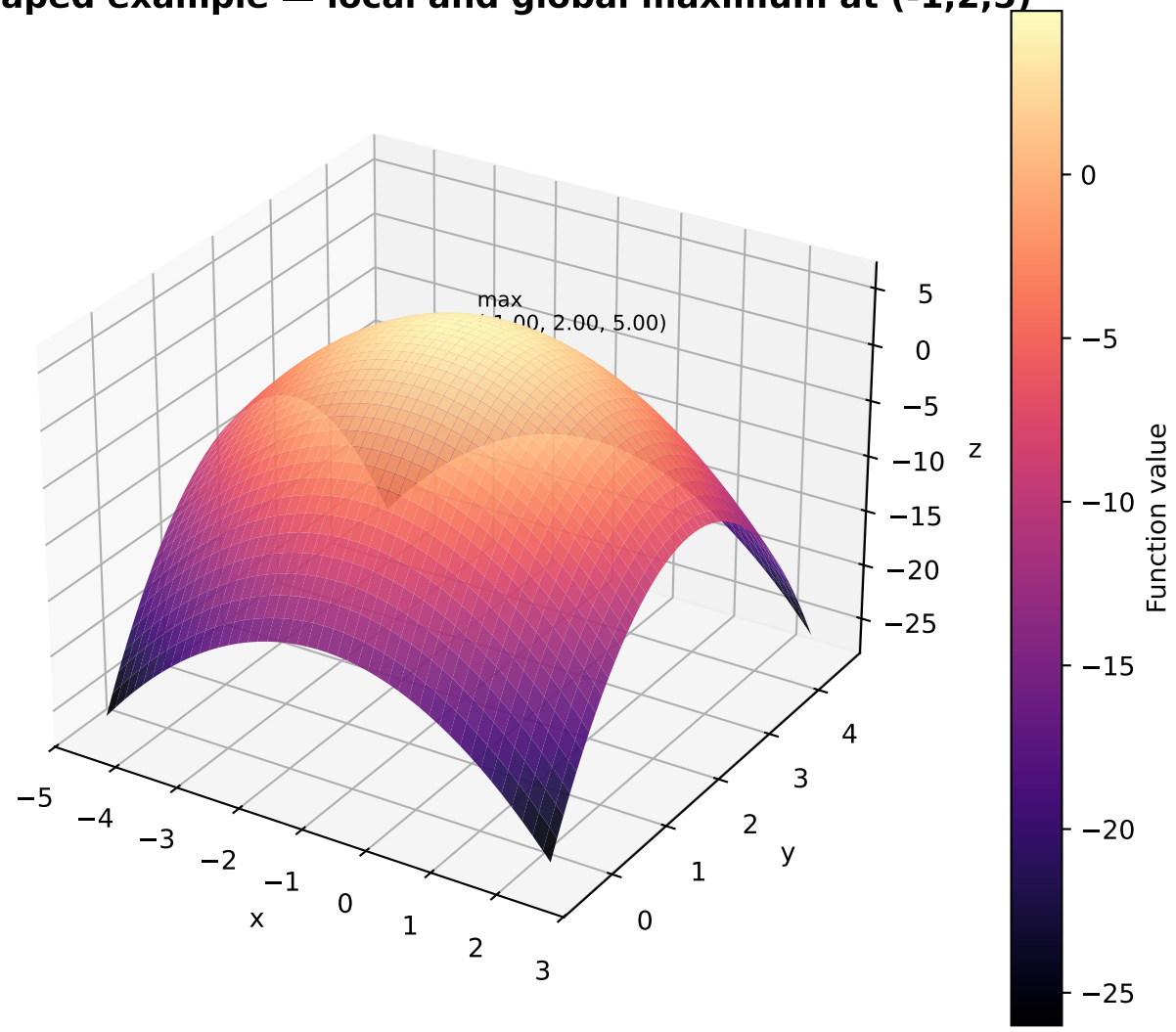
ed design box — absolute minimum at (1,2,0), absolute maximum at (4,5,18)



Key point(s): abs min, abs max
Cap-shaped surfaces typically correspond to maxima.

$$M(x, y) = 5 - (x + 1)^2 - 3(y - 2)^2$$

Cap-shaped example — local and global maximum at (-1,2,5)



Key point(s): max
Cap-shaped surfaces typically correspond to maxima.